CHILL DIVISION

Chill Division Site Design Guidelines

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CHILL DIVISION

Cultivation Facility

Site design guidelines

Introduction

These ***Site Design Guidelines*** work together with the ***Chill Division Security Procedures*** and the ***Chill Division Cultivation SOP*** documents. You will notice that the first chapter is similar across both the Site Design and Security documents. This is because the information is crucial for both planning your site and keeping it secure. A licence application should include all 3x documents.

While regulations allow for indoor, outdoor, and glasshouse cultivation, this document focuses on indoor cultivation. This is because growing indoors is the most reliable way to consistently meet the NZMQS microbial standards for quality. These guidelines are flexible enough to work for a fit-out in a rural shed, or a warehouse in a suburban industrial area. The main goal is to help you cultivate high-potency THC-dominant flower that will be sold as a final dried-flower product.

The design of your facility is important. A well-designed space helps your staff do their jobs well, keeps any smells from bothering your neighbors, and ensures all cannabis materials are safe and secure. An intentionally designed facility is also more efficient. It helps maximize CO2 injection, improves insulation for your air conditioning to run smoothly, and minimizes the need for air exchanges. With careful planning, you can dedicate up to 50% of your total floor plan to flowering canopy space. An optimized room also provides easy walking access to all your plants, ensures seamless drainage, and gives you consistent control over the environment, making for a reliable cultivation experience.

This document contains many recommendations. Many of these can be customized to fit your specific needs, preferences, and budget. With that said, we *strongly* recommend using Home Assistant. It is a powerful, open, and flexible automation platform that works with all the common equipment a master grower might need. It is by far the easiest way to integrate and unify a variety of different products / platforms necessary to maintain a successful grow.

Most importantly, this document is here to give you a clear understanding of what "reasonable" precautions look like as you design your facility. You will need to provide more specific details to the Medicinal Cannabis Agency during your application process. However, if you follow these guidelines, your facility will likely exceed these minimum requirements, which will help your application be successful. Please note that this document covers the setup and build of your facility, not the specific details of cultivation.

All terminology in this document follows the standards set out in RFC 2119, and we assume you are familiar with it.

1. Building security requirements

These procedures will aid in selecting a premise for lease to cultivate in, along with ensuring that the appropriate amendments to the facility are made to ensure the safety of cannabis.

When searching for a lease, a facility owner should be mindful of the amount of time the lease is likely to last, and so seeking a 4-5 year lease term, with 2+2 ROR (Right of Refusal) is often desirable. If a facility owner is seeking to have 50-60sqm of canopy space, a 160-200sqm warehouse lease is likely going to be sufficient, with further optimizations able to be had by an experienced owner.

The Landlord of this warehouse must be willing to provide a letter to the Medicinal Cannabis Agency to include with your licence application. It must state their approval of your undertakings at the location, so it pays to be up-front with your landlord, and engage a regulatory consulting expert who can assist with ensuring a successful lease.

It is also worth noting that in cases where some of these guidelines are unable to be followed, the totality of the security arrangements will be assessed by the Medicinal Cannabis Agency, and there may be flexibility in some instances. However, the assumption of oneself being “the exception to the rule” is not an optimal place to start, but rather a place whereby a regulatory consultant may be able to offer ways to make a location suitable even if it does not match 100% of the criteria from the outset.

During build-out, and before the Medicinal Cannabis Agency’s site audit, Responsible Personnel should also notify the New Zealand Police of the proposed location and the nature of activities to be conducted on-site — see the Chill Division Security Policies & Procedures for details.

Location

1.1 The premise should not be within 120m of any residential properties. This is the minimum distance to aim for; where a prospective site cannot meet it, the location should be reviewed with a regulatory consultant on a case-by-case basis, and may still be accepted at the Medicinal Cannabis Agency’s discretion.

1.1.1 The premise should ideally have at least a 150-200m radius around the facility location whereby there are no other residential properties, unless the house is that of the owner of the cultivation facility or similar.

1.1.2 The premise should not be within 200m of a school, kura, kindergarten, preschool, highschool, or other place of education.

1.1.3 The premise should ideally be in a built-up industrial area.

1.1.3.1 A commercial area may be considered but is not recommended due to the possibility of elevated foot-traffic from pedestrians and the general-public.

1.1.4 Rural residential properties may be considered as a suitable location for a cultivation facility, such as a greenfield site, but with the expectation that cultivation will take place indoors. This indoors may be a shed, a container, or a constructed premise.

1.1.4.1 Any site should still ensure there is at least 120m from the location of the desired cultivation facility building, and any neighbouring residential homes, as with item 1 above.

1.1.5 The building may be adjacent to other industrial units in a “light industrial complex” type of situation.

1.1.5.1 Thick concrete walls or other secure framing should be in-place between the units.

1.1.6 Where a standalone unit is sought for cultivation, such as a shed or individual warehouse, there must be two barriers to entry at all times to any room where cannabis materials will be held. As such a fence / gate may be used as one of these barriers, however it must be able to stand on its own merit and should be a suitable distance from the premise.

1.1.7 Consideration should also be given to the layout of any pre-existing warehouses, as certain rooms may be unsuitable for use for cultivation aspects due to existing doors / walkways within the facility.

1.1.7.1 This often includes a “lunchroom” on the externally facing wall which also has an adjacent roller door. Due to the roller door being a single barrier to entry, although there may be two doors any personnel would normally enter through via the lunchroom, additional internal barriers to entry would need to be considered

General security principals

The majority of these requirements will be fit-out requirements that the facility owner will have to implement, and are unlikely to be available in any pre-existing setup. These principals will guide the design of the facility.

1.8 There should be no external signage, logos, emblems or markings anywhere on-site that relates to the company, or the nature of the business being conducted on-site

1.8.1 If required as part of body corporate signage etc, a decoy name should be used, or that of a separate / unrelated business.

1.9 Generic and unidentifiable alarm / security stickers are permitted for the purpose of deterring diversion attempts / unauthorized access.

1.10 PINs are not to be written down anywhere, nor any identifying marks made on the key fobs that would pertain to the company (such as name or address) or any other such references to the site / location.

1.11 Keys are not recommended for physical access to any locations of the site due to easy cloning and copying. Where no other alternative is available (such as for a gate / fence), keys may be used and should be marked as "**DO NOT COPY**".

1.11.1 For any padlocks, a tumbler-PIN lock should be utilized if no other alternatives can be used, instead of a key-based lock.

1.12 Access to any area of the site which may contain cannabis biomass must employ a method of *2-factor authentication* (Two separate factors of authorization) in order to reach any room where cannabis materials may be. This means one door may require a fingerprint whereas another requires a PIN.

1.13 2-factor authentication methods may include but are not limited to a combination of a PIN + fingerprint, PIN + key fob, fingerprint + key fob, PIN + NFC authentication method or other similar security arrangement. This ensures that should a Key Fob / cellphone be lost, stolen, or go missing, its use would not permit any unauthorized access to the site.

1.14 An alarm must sound in the event the external personnel doors remain open for a period of more than 60 seconds. A siren must sound inside the facility, acting as a deterrent to any attempted diversion, should a break-in be attempted.

1.14.1 Responsible Persons will also be automatically notified electronically of the security breach so that security footage can be reviewed and appropriate remedial actions can be taken including calling of the Police if required.

1.15 Ensure that unauthorized individuals, including guests and other personnel, are unable to view the entry of the PIN code on any keypad.

1.16 Ensure that unauthorized individuals are unable to gain access to the cultivation facility by tailing authorized personnel through a closing door.

1.17 Personal belongings should be left off-site where possible, such as in your vehicle. Any bags or other items taken into the secure areas may be searched before you exit the building.

1.18 The site must be operated in a manner such that smell does not escape to bother nearby neighbours or businesses. This may involve operating the facility at a **negative pressure** and / or **carbon filtration** for any venting of air.

1.18.1 This may also include internal scrubbing of air with carbon filtration.

1.18.2 HEPA / MERV filtration is not an acceptable method of odour control, as the filters are only designed to capture dust and other particulate matter, and only carbon filtration is acceptable for scrubbing odour.

1.19 To further combat odour escaping, all rooms where cannabis materials are held should be sealed with a silicon-based beading (Such as Sikaflex AT-Facade) along any gaps in wall-panels, joins, edges, skirting boards, and around drainages. This will minimize odour escape while also maximizing CO2 injection efficiency.

1.19.1 Care should be taken where any electrical / data cabling may enter the room, as well as plumbing and CO2 related penetrations.

1.20 Similarly, doors to those rooms should also be fitted with an edge / automatic door bottom sealer, such as Ravenseal RP10Si frame-seal / RP8Si door bottom automatic sealer, to prevent further odour escaping or CO2 leaking.

1.20.1 Alternatively where doors can be retrofit / replaced, a hermetically sealed door should be used.

1.21 Documents such as SOPs, Floorplan etc must always include a version number. This should be in the form of Semantic Versioning (Major.Minor.Patch) format as defined by the Semantic Versioning 2.0.0 specification (semver.org), or date-based versioning as described in ISO 8601 (YYYYMMDD).

1.22 Any floorplan provided to the Medicinal Cannabis Agency must label each internal room with its purpose (Flower1, Flower2, Nursery, Drying etc). Each physical room must then be labelled with a matching sticker / poster, on or directly adjacent to its door, so that the physical labels always align with the floorplan.

1.22.1 The locations of free-standing equipment relevant to security or odour control — such as carbon / HEPA filters — should also be noted on the floorplan.

Physical Security Arrangements

1.24 If there is a fence acting as a barrier to entry, it must be sufficiently far from the other barrier to entry (such as building exterior) that it is able to stand on its own merit as a defense mechanism.

1.24.1 The fence should be of sturdy construction, such that it would stand up to a ram attempt from a vehicle.

1.24.2 The fence should be of a climb-proof construction, e.g. flat-faced wooden planks vertically aligned with a minimal gap between them, or vertical metal bars, but not of diamond-mesh that can be used as a hand / foot grip.

1.24.2.1 Depending on the height, it may be suitable to also install barbed wire along the top of the fence.

1.24.3 The fence must not be able to be jimmied up from the bottom, such as with a spade or crowbar. This is especially important if it is a metal fence that doesn't have a thick and solid bottom.

1.24.4 The fence must be of a decent height such that it is unable to be leaped over, with a minimum of 1.8m but ideally 2m+

1.24.5 The fence must not be an excessive height off the ground such that somebody could slip underneath it, such as 120mm or less.

1.24.6 Any gate must be able to automatically swing shut and re-lock

1.24.7 Gate access must also follow the same access control as the rest of the facility, and as such maglocks are usually recommended.

1.25 All windows on any ground floor should be barred or boarded with plywood such that in the event of an unauthorized person attempting to gain entry to the site, they would still be unable to do so.

1.26 If the roller door does not have an external bollard, (e.g. there is an internal bollard), then any externally accessible roller doors must be latched internally in such a way to prevent jimmying open from the outside, such as with a crowbar.

1.26.1 This may be in the form of an internal latch, or maglocks appropriately spaced along the length of the roller door.

1.27 All roller doors must also have some form of additional internal or external barrier, such as a bollard or ram-arm.

1.28 If there are external glass doors, these must also be barred internally.

1.29 All external doors, as well as any internal doors to rooms that may contain cannabis materials, must have an automatic closing mechanism installed.

1.30 All external doors, as well as any internal doors to rooms that may contain cannabis materials, must have an automatic locking mechanism.

1.30.1 Doors with locks must relock immediately upon closing.

1.31 All PIN locks must automatically lock the keypad after a maximum of 5 incorrect attempts in a row, and must not unlock for a period of 3 minutes, to deter any brute force attempts.

1.32 Locks must remain operational at all times. A battery UPS must remain in place to maintain the things such as mains-powered maglocks in the event of a power failure.

1.32.1 All digital locks must have an expected battery life of 12+ months, with the intent that even at critical battery levels they will remain operational for a period of no less than 1 week. This is to ensure that in the event of a fire / power outage the locks will always permit staff to exit the facility.

1.33 All external doors, and any doors to rooms that may contain cannabis materials, must be fitted with a door-state sensor so that door open / close events are logged with a timestamp, and so that alerts (such as the alarm for doors remaining open for more than 60 seconds) can function.

1.33.1 Zigbee contact sensors are the most common and practical choice for this. Alternatively, wired options such as hall-effect sensors or limit-switch style sensors, attached to a PoESP32 or Atom Lite, may be used.

1.33.2 Roller doors should also have state sensing, so that an unexpected opening outside of business hours triggers an alert.

1.34 If there is any skylight, it must be frosted such that if work is being undertaken on the roof (e.g. Bodycorp repainting a roof) they are unable to see inside and ascertain the undertakings within.

1.34.1 Skylights should be painted entirely where possible, to avoid light bleed, especially in cultivation rooms where photosensitive plants may be.

1.34.2 If possible, a fake secondary ceiling should be constructed on the inside, both as an odor control measure, allowing for insulation, and further obscuring the room contents.

1.35 Cameras must be installed in every room that will contain cannabis materials, such as grow rooms, drying spaces and disposal areas.

1.35.1 Cameras do not need to be in rooms that will not contain cannabis materials, such as lunchrooms or changing areas.

1.35.2 Cameras may be installed in hallways or other functional areas where cannabis movement between rooms may be.

1.35.3 Larger rooms will likely require two or more cameras, with the primary camera having a close / unobstructed view of the room entrance, and others showing a broader perspective of the room to detect intruders / movement.

1.36 Care should be taken when selecting a camera for any room that will have growing plants, the night-vision using far-red (iR) can interfere with plant photoperiods where the spectrum falls inside the 400-700nm PAR range.

1.36.1 Cameras such as the Reolink RLC-510A or RLC-810A are recommended due to having infrared LEDs that are approx 830nm, and should not interfere with a plants photoperiod and ability to “sleep” at night.

1.36.2 Cameras must be cabled and PoE-powered — battery-operated cameras are never acceptable. Any camera selected must support at least 1280x720 resolution at 2fps or better, continuous 24/7 recording (motion detection may supplement, but never replace, continuous recording), and automatic IR for clear recording in darkness.

1.37 Cameras must also be installed outside, facing in every possible direction of approach for a would-be intruder.

1.37.1 For standalone buildings this will likely be a camera on each corner facing a 90-degree rotation from the previous camera.

1.37.2 For facilities part of a “block” of buildings, this will likely be the front in two directions with the cameras “crossing over” view of each other.

1.38 Where there is sufficient space above any fake-ceilings, cameras must also be installed in order to offer an early detection of movement for any attempted intrusion via the roof / ceiling.

1.38.1 This should ideally be two cameras that cross over vision, rather than just a single camera

1.39 An NVR should be selected that has the capacity to hold data for at least 30 days.

1.39.1 The exact capacity will depend on the number of cameras, and the amount of movement those cameras observe, but a good rule of thumb is 3TB for 2 cameras (When recording with a video bitrate of 4096kbps) for a months storage.

1.39.1.1 For example, if a facility has 12 cameras, then there should be 18TB of storage made available.

1.39.2 Where possible, a RAID-5 or RAID-1 setup is recommended for data redundancy.

1.40 A Duress Button should be installed near all main entrances to the premises, such that if a staff member is forced to open the building, under duress or threat of physical harm / violence, they can discretely push this button upon entry.

1.40.1 Having them situated near primary lighting is considered to be an optimal location. However, these buttons should be unmarked so that there is nothing to indicate that they have triggered a silent alarm notifying other responsible personnel etc.

1.40.2 A wireless Zigbee button such as a Xiaomi / Aqara Mini Button is ideal for discrete and flexible placement.

1.40.3 Alternatively, a Mini Dual Button unit, or Mechanical Key Button, attached to an Atom Lite or PoESP32 may be utilized to the same effect

1.41 A standalone controlled-drug safe / vault is not required, as the Medicinal Cannabis Agency holds overriding jurisdiction and discretion over storage arrangements. Instead, the facility owner must ascertain the most secure room within their facility to act as the secure storage area for packaged product and any biomass staged for dispatch, and note this room on the floorplan provided to the Medicinal Cannabis Agency.

1.41.1 Most facilities will find their drying or processing room to be the most suitable choice, as these rooms already meet the access control, surveillance, and environmental requirements for holding cannabis materials.

1.41.2 The room selected must meet all of the same requirements as any other room holding cannabis materials, including 2-factor authentication for access, camera coverage with a close / unobstructed view of the entrance, and automatic door closing / locking mechanisms.

1.42 Where the facility has a roller door, the loading bay area should be designed such that a delivery vehicle can be reversed flush up against the opening and the roller door area obscured, so that loading / unloading of any goods, plants, or cannabis materials cannot be observed from outside.

1.42.1 Almost every facility will only have a small area by the roller door that won't fit a whole vehicle inside — this is acceptable to the Medicinal Cannabis Agency, provided the vehicle reversed flush against the opening obscures visibility during loading / unloading.

1.42.2 The loading bay should have its own camera coverage, in keeping with the camera requirements above.

1.42.3 This area will also serve as the reception point for large deliveries, new plant introductions, and contractor vehicles / tools, so consideration should be given to keeping it clear of permanent fixtures.

2. Grow room

The grow room design is repeatable for both separate vege / flower, or combined rooms. In order to maximize utilization, many growers in newly designed facilities are opting to "vege in-place" and instead of having a dedicated vegetative grow space (which would be used for only a short amount of time) to use it as flowering canopy space.

For a loose rule of thumb, in a 200sqm warehouse, you'll likely have:

• 20sqm for any loading bay area

• 20sqm for bathroom / kitchen

• 25sqm for mothering / nursery area

• 35sqm for hallways and other wasted space

• 20sqm for drying area

• 80sqm split between two flower rooms

Based on these estimates you would likely have approx 40-50sqm of flowering canopy space.

Overall layout

2.1 The grow room should be designed with as little additional items inside it as possible. Storage of spare parts for filters, fans, tubing etc should be stored outside of the grow room.

2.2 Thoughtful consideration should be given to physical access to every individual plant, including walkways, drain table width, and other obstructions.

2.2.1 Isles and walk spaces should be a minimum of 800mm wide, as is a standard door width. 1200mm is preferable

2.2.2 Isles should also exist around the exterior of the room so plants can be accessed from more than one side — see the Tables, isles and plant density section (item 79) for table placement and access requirements, including the moveable-table exception.

2.3 Rooms must be insulated to at least an R2.0 value on the walls / ceiling. Where insulated chiller panels are used as the wall / ceiling lining (see §2.10), the panel itself provides this insulation; the guidance below applies to any non-panel build-up, such as the wall behind ACM board.

2.3.1 An R3.0 value or higher is recommended for the ceiling, with some facilities opting for R4.5 where possible.

2.3.2 Insulation may be Pink Batts, GreenStuf, Stonewool or EPS/XPS.

2.3.2.1 Consideration should be given to the use of EPS/XPS and how this will impact the facilities' capacity to obtain insurance, even if fire retardant EPS/XPS is utilised.

2.3.2.2 Under the New Zealand Building Code (NZBC Clause C), stonewool panels achieve a Group 1-S or Group 1 fire rating (the safest classification possible). Stonewool is preferred for this reason.

2.4 The floor should be covered with a self leveling compound, thin layer of resin, fibreglass or other such coating to both protect any concrete / plywood underneath, as well as to prevent VOCs from leaching from concrete. This is especially important in leased properties that may have previously been utilized by mechanics etc.

2.5 Where possible, the floor should be elevated and / or an additional secondary floor constructed such that any spills, overflows or floods can easily be drained away without the use of any additional equipment.

2.5.1 It is recommended that a Wet/Dry Vacuum Cleaner remain on standby at all times, outside of the grow room, in case of an emergency clean-up.

2.5.2 Battery operated Wet/Dry vacuums are a good option for maximum portability and quick deployment in case of emergency.

2.6 Ideal heights for a room are 3.0 -> 3.4m. Shorter rooms are viable, even as small as 2.4m, but additional consideration needs to be given to plant height during flowering. Taller rooms are also still viable, but air circulation / heating / dehumidification etc can become problematic for rooms that are too tall.

2.6.1 "Double stacking" layers of plants is not recommended, specifically not for a first facility, even where the height of a room may initially look to permit it. Instead, a second false-ceiling should be constructed and insulated around it.

2.7 Seal the grow room for odour and CO_2 containment as required under §1.19–20. In addition to the beading and door seals specified there, the following apply to the grow room specifically:

2.7.1 A seal should be applied around any door framing, between the frame and the wall.

2.7.2 Where any fertigation, plumbing or drainage enters / leaves the room, sealant should also be applied.

2.7.3 Advice from an electrician should be sought regarding any power or ethernet sockets. The faceplate itself should not be sealed, however it is worth seeking advice regarding sealing under the faceplate.

2.8 The door to the room must be sealed as specified in §1.20 — hermetically, or with a flexible silicon frame seal such as the Ravenseal RP10Si, together with an under-door automatic retracting seal such as the RP8Si where there is any gap between the door and the floor.

2.9 Regardless of the lining chosen, all internal wall and ceiling surfaces must be non-porous and wipeable. Insulated chiller panels and ACM board both satisfy this by default; where a porous substrate such as plywood is used it must be sealed with a non-porous coating such as fibreglass.

2.10 Walls and ceilings should be lined with insulated chiller / coldroom panels as the base expectation. These are sandwich panels with a rockwool / stonewool core between two skins — aluminium-skinned rockwool panels are the preferred baseline, and Colorsteel / steel-skinned rockwool panels are equally acceptable. A single panel delivers insulation, a non-porous washable surface, and a Group 1-S / Group 1 fire rating together. Where insulated chiller panels are not used, Aluminium Composite Material (ACM) board may be used as an acceptable secondary option; ACM provides a non-porous, wipeable surface but is not itself an insulating panel, so it must be installed over a separately insulated wall build-up (see §2.3) to meet the insulation expectation.

2.10.1 Consideration needs to be given to the thickness of the panels, as a 6-7m span requires 150mm thick stonewool (varies depending on skin thickness etc).

2.10.2 Shorter spans may be able to use 100mm or 75mm thick walls, though this will impact insulation.

2.10.3 Alternatively, the panels may be anchored to the primary roof structure of the building. Note that doing so changes the internal structure into a permanent fixture of the building (rather than a free-standing internal room), which will likely require building consent — seek advice before proceeding.

2.11 When designing the layout of the floorplan, consideration should be given to the length for “Dead end to open path” as referred to in the C/AS2 Acceptable Solution for Buildings other than Risk Group SH (Table 3.2). 25 meters is usually the target for small cultivation facilities classified under Risk Group WB, and avoids the requirement of a sprinkler system installation.

2.11.1 Where unsure, seek advice from a Fire & Safety consultant.

Sensors / microcontrollers

Information on setting up and example yaml configuration for the most commonly utilised M5Stack microcontrollers and sensors is available on our github:

https://github.com/Chill-Division/M5Stack-ESPHome

Note: M5Stack have marked the ENV.IV unit as End of Life, and have returned to producing the ENV.III. Both are functionally equivalent for the purposes of this document (temperature, humidity, and barometric pressure), so they are simply referred to as "ENV Sensor(s)" throughout, and whichever unit is currently available may be used.

2.12 All sensors must be physically cabled-in wherever a cabled option exists, as opposed to WiFi or Bluetooth. Where no cabled option is available, only then may WiFi or Bluetooth be considered.

2.12.1 The use of the M5Stack PoESP32 should be considered as the gold-standard for microcontrollers.

2.12.2 The majority of a cable length to a sensor will consist of the ethernet cable, from the patch panel / wall-jack, to the PoESP32.

2.13 Each room should have a device such as an ESP32 Atom Lite to act as a Bluetooth Proxy, and repeat bluetooth signals back to your automation server.

2.13.1 This may also be used for in-room device presence detection, through the likes of Bermuda BLE Trilateration.

2.14 Sensors will ideally connect through a Grove port, back to the PoESP32.

2.15 Consideration must be given to the maximum length of cables for i2c devices, with 1m cable runs often being near the limit, especially where splitters are utilised.

2.16 An SCD4x sensor should be used for CO2 sensing, as well as VPD calculations.

2.16.1 Multiple sensors should be used, with at least one for each end of a room (lengthwise).

2.16.1.1 The sensors should be placed at least at each end of the tables/room.

2.16.1.2 Ideally, sensors will be placed in a quincunx pattern on the floorplan. One in each corner of the growroom and one in the middle.

2.16.2 Where CO2 injection takes place above the canopy, “raining down” on to the plants, it is recommended that the sensor be hung just beneath where the canopy is expected to be.

2.16.3 It is recommended that `automatic_self_calibration` be set to “false”, especially where devices are not regularly exposed to 400ppm levels (such as in a sealed grow room environment).

2.16.4 Additional sensors may be placed in strategic areas around the grow room to ensure an even spread of CO2 throughout.

2.16.5 Sensors should calculate the VPD on the PoESP32 device directly, with the calculation:

```
lambda: |-
return (((100 - id(humidity).state) / 100.0) * (0.6108 * exp((17.27 *
id(temperature).state) / (id(temperature).state + 237.3))));
```

2.17 An ENV Sensor should be used for Temperature, Humidity, and VPD calculations where CO2 sensing is not required.

2.17.1 For example, a CO2 sensor may be run off a 1-to-3 Grove Splitter and hung under-canopy, while an ENV Sensor off the same splitter is situated above the canopy, providing multiple datapoints for VPD. This will also help to minimize the number of PoESP32 microcontrollers required. However, consideration should be given to i2c address collision as well as maximum cable lengths.

2.17.2 ENV Sensors can be useful to determine barometric pressure, and ascertain if a grow room is positively or negatively pressurized. Be aware that grow rooms must either be fully sealed, or operated at a negative pressure. The specific pressure differential targets and ongoing testing expectations are set out in the Chill Division Security Policies & Procedures document.

2.17.3 Multiple sensors should be situated in a variety of places around the grow room, both at canopy-level but also in other non-canopy areas to provide a bigger picture of the airflow within the room and help avoid hotspots.

2.18 A PM2.5 sensor may be utilized to observe particulate matter, such as a PM2.5 Air Quality Kit (PMSA003 + SHT30). This is one of the few sensor examples where WiFi may need to be used.

2.18.1 This unit may be placed anywhere within the room, however its use will likely be limited to timeframes outside of cultivation due to Humidifier droplets causing them to give false readings.

2.18.2 Where there has been no Humidification for 12 hours (such as during the lights-off photoperiod), the PM2.5 and PM1 values should be $<=1\mu g/m^3$.

2.19 ENV.Pro units may be used to observe VOCs and IAQ values.

2.19.1 These values have not had any direct observed impact on cultivation / microbial levels, so the use of an ENV.Pro may be used to simply gauge a baseline prior to any plants being introduced.

2.19.2 It is recommended that an ENV Sensor be used instead in most instances.

2.20 A smoke sensor such as a “Grove - Gas Sensor(MQ2)” should be used to monitor for any smoke. These can trigger all-of-building alerts, as well as remote monitoring alerts.

2.20.1 Because smoke rises, these should be placed strategically near the roof, ideally in multiple locations.

2.20.2 Additional standalone or “dumb” smoke sensors with alarm / sirens should also be used in the grow room. Smart sensors should not be used as a complete replacement.

2.20.3 The sensors should be calibrated if using the analog pin, by burning a test-flame nearby such as a small piece of paper. Make notes of levels for when there is vs is not smoke, as part of calibration for any alert levels.

2.20.4 Alternatively if using the D0 pin for binary detect/non-detect a test should take place when installing them to confirm accuracy. This is due to the units having a longer “warm up” period if they haven’t been used in several weeks as is often the case during initial manufacturing / shipping.

Air conditioning

2.21 Air conditioning must be employed, such as a mini-split or ducted system to control the temperature inside the grow room.

2.22 Advice from a specialized air conditioning installer should be sought to determine capacity requirements.

2.22.1 The expectation is that the air conditioning units will only maintain temperature, and not dehumidification.

2.22.2 Consideration should be given to both the area of the room in m3, as well as offsetting the BTUs caused by LED grow lights, undercanopy supplementary lighting, dehumidifiers, oscillating fans, ducted / filtration fans, and any pumps.

2.22.3 Air conditioning should ideally be over-specced by 20%, as this will minimize the impact of any future changes around dehumidification, lighting upgrades, additional fans etc

2.23 Air conditioning may need to be controlled using WiFi through an intesishome_local device or similar. Some may be controllable through RS485, however the integration with an automation system should be tested prior to committing to a full fit-out.

2.24 Air conditioning may be setup with the use of external temperature sensors for controlling it, such as with the ESPHome IR Remote Climate. This may provide a better representation of the actual temperature in the room compared to what the units themselves may detect, however this is not a requirement.

2.24.1 Additional considerations need to be given to the placement of any iR sensors if this is to be implemented.

2.25 High wall units will often be placed on the short side of the room, facing lengthways, to prevent short-cycling.

2.26 High wall units should be setup with continuous gravity-fed drainage in mind, without the need for pumps where possible.

2.26.1 Consideration needs to be given to the area physically underneath the high-wall unit, where units may leak or drip water. Ensure there are no electronics or other things on the wall or floor underneath the unit.

Ventilation

2.27 Ventilation is the primary manner in which a crop will pass or fail microbial testing, and a well thought through ventilation system is crucial for any grow room.

2.28 The expectation is that as a sealed room, there is no immediate "need" to constantly vent air out, though some growers may want to do this at night etc.
However, there is no evidence to suggest it is detrimental to plants if this does not take place. By not venting out and doing a complete air-exchange, the grower can save on the resources required to heat / cool, humidify / dehumidify, and CO2 enrich the air in the grow room.

2.29 Airflow will usually go through 3 stages of carbon filter -> HEPA filter -> inline or centrifugal fan, in that order. This may even be standalone on the floor with the base being the carbon filter, a HEPA filter sitting on top of that, and then the fan sitting on top of that.

2.30 Inline fans have been found to perform best overall instead of centrifugal fans and should be considered as the baseline.

2.30.1 Where remote control of fans is required, AC Infinity 69 Pro+ controllers are known to work with Home Assistant. The AI-models are currently "read-only" and do not offer control through the integration.

Internal air circulation

2.31 Internal air circulation and the flow of air throughout the room should ensure a consistent level of CO2, humidity and temperature throughout. Because regular / constant venting out is not the recommendation, a clean and stable internal atmosphere should be maintained.

2.32 While some may view "hot spots" of elevated / reduced temperatures / humidity / CO2 as inevitable, a well ventilated room should minimize the degree in which these impact the plants.

2.33 A single 200mm fan operating at approx 500CFM or 850m^3/hr should be allocated to approx 85 -> 127m^3 of airspace within the room, to allow for a full air movement every 6 to 9 minutes.

2.33.1 Too high of a flow can be problematic, and may cause inadvertent wind-burn on plans

2.34 Larger fans may be used, however 200mm is usually an ideal size for well distributed airflow as opposed to high volumes in specific areas as with a 300mm fan.

2.35 Wind socks should be employed to distribute the air throughout the canopy, causing a very slight "rustle" of the fan leaves. This should be at approx 0.4m/s to 1.2m/s.

2.35.1 Wind socks can be affixed to either 150mm or 200mm fans.

2.35.2 The wind socks should be run down the length of the drain-tray undercanopy, aiming upwards at the canopy.

2.35.2.1 Care should be taken to ensure it is not alongside undercanopy lighting which may cause it to melt or degrade.

2.35.3 Wind socks may also be used above the canopy, for example between the ceiling and any lights.

2.35.3.1 Caution must be exercised when doing this to ensure that the wind sock is not too close to either ceiling lights or over-canopy grow lights which may cause it to melt or degrade.

2.35.4 Oscillating fans aimed at the canopy are not specifically required where wind-socks are used.

2.35.5 Ceiling fans that lift the air up from the floor spreading it up and outwards may be used, and can be an ideal way to move hot air away from above lighting fixtures.

Air scrubbing

2.36 Air scrubbing is used to remove both particulate matter as well as odour from the air inside the grow room through the use of **both** HEPA and Carbon filters.

2.37 Carbon filters should be used to remove odour from the grow room, but they do not remove particulate matter, such as yeast / mold / bacteria.

2.38 HEPA filters should be used throughout the grow room to remove particulate matter / bacteria from the air, but they do not remove odour.

2.38.1 Although it may not be practical to install a HEPA filter onto all fans, such as under canopy windsock fans, it should be prioritized for primary air recirculation so that all air passes through a HEPA filter with at least 6 changes per-hour (every 10 minutes). This may mean attaching them to other recirculation fans

2.39 Where a HEPA rated filter is not available, a ULPA filter or a filter with a MERV rating of 16+ may be used. A MERV rating of lower than 16 should not be used, and a better filter sourced instead.

2.39.1 This will directly impact the grow rooms ability to meet the NZMQS microbial standards and priority should be given to ensuring filtration standards are met.

2.40 Carbon filter efficacy varies greatly, and oftentimes the cheapest carbon filters will not adequately remove odour. Use of well renowned name-brand filters, such as AC Infinity or Mountain Air is recommended.

Venting air out

2.41 Venting air out of the room is not specifically required, and a grow space that is fully-sealed in every way possible is strongly recommended. However, some growers may opt for a nightly air exchange to avoid unwanted effects such as an ethylene build-up, or using it as a chance to reset the room environmentals.

2.42 When air is vented out, care must be taken to ensure this is done through a carbon filter to capture any odour.

2.42.1 More regular replacements of any vent-out carbon filters may be necessary in order to ensure odour is not a nuisance to neighbours.

2.43 Any opening will naturally compromise the room seal in some manner, so if a vent is desired then a backdraft damper should be utilized, and a cowl on the outside of the room. This would have a carbon filter inside of the grow room, then a fan, and then an external roof cowl.

2.43.1 Electronic dampers are recommended (rather than spring-loaded), with a foam seal to prevent accidental leakage.

Filtering air in

2.44 The use of an airbox / antechamber to filter air prior to going into the grow room should be used. For example, filtration of a mantrap room, or an adjacent sealed hallway.

2.44.1 The airbox / antechamber must also be fully sealed with silicon.

2.44.2 The airbox / antechamber should operate at a positive pressure, pushing clean and filtered air into the growspace.

2.45 The air coming in should pass through a HEPA filter to remove microbial contamination, and a carbon filter to remove VOCs.

2.46 Additional sensors should be used in any airbox room to monitor barometric pressure.

Humidification

2.47 Humidification is essential to maintain a tightly controlled VPD. It is far easier to humidify air than it is to dehumidify it.

2.48 As such a dehumidifier will often have a broader setpoint (Ex. 65% +/- 3%), with the intention of the setpoint being drier than necessary to achieve an optimal VPD, and a humidifier used to tighten the variance.

2.49 Smaller / standalone humidifiers are *not recommended*, as manual top-ups can easily be overlooked and missed, resulting in fried plants. Humidifiers should always be plumbed into filtered water, with possible exceptions being made for a VPDome or similar style cloning setup.

2.49.1 UV treatment should be used for any water being fed into a humidifier.

2.49.2 An additional secondary UV treatment or filtration of water going to a humidifier may be desired, as the water that passes through the humidifier will effectively be "applied" to the canopy, along with any microbial growth / bacteria in the water. Details are included in the Irrigation System section of this document.

2.50 A large 320w 5.5L/hr humidifier should be sufficient for approx 90-100m^3 of airspace.

2.51 Care must be taken to ensure that the humidifier is physically situated in such a way that the droplets do not immediately come into contact with plants, electronics, or other equipment.

2.51.1 Physical orientation being near dehumidifiers, aiming towards the canopy, is often suitable as either the humidifier or dehumidifier will be going at any given time, both need not be operating concurrently.

2.51.2 Care should also be taken to ensure that it is not directly in front of any fan (inline or oscillating) such that it would cause the droplets to condensate and form, or be sprayed around the room.

Dehumidification

2.52 Dehumidification becomes even more crucial in an enclosed / sealed environment, as the plants will transpire and release a significant amount of humidity into the air.

2.53 Although not all of the water from fertigation that a plant consumes will be thrown into the atmosphere, a significant amount of it will be, and this will be significantly concentrated during the lights-off photoperiod. As such, thoughtful consideration should be given to dehumidification capacity.

2.53.1 It is better to have more dehumidification capacity than not enough.

2.54 Canopy size vs PAR levels vs a plants' capacity to yield well will be the primary basis for your calculations. A loose rule of thumb should be 10L/day of dehumidification per-sqm of canopy.

2.54.1 For a canopy of ~20sqm, assuming based on 4x plants per-sqm (80 total), a loose expectation would be that you would need to soak up 200L/day if each plant is given a peak of 2.5L. The plant absorbs part of that, while part goes into runoff as well, however that amount needs to be predominantly absorbed while the plant sleeps (which is only half of the photoperiod).

2.54.2 For a canopy of 50sqm, assuming 9x plants per-sqm (450 total), watering each plant a total of 1.2L/day, it would still match the expectation of needing 500L/day of dehumidification capacity.

2.54.3 This will also vary greatly depending on the genetic morphology, plants overall yield capacity, density of plants, defoliation techniques and more.

2.55 Having ample physical space for an additional dehumidifier to be installed in the future is always wise.

2.56 Consideration should also be given to the rated efficacy of a dehumidifier, and optimal operating conditions, be it 80% at 30C or 60% at 26C for example.

2.57 The dehumidifier units should ideally be situated higher up in the grow room.

2.58 The dehumidifier units should also expect to continuously gravity drain any recovered water, as small pumps inside units are prone to failure.

2.59 Consider redundancy, such as 2x 150L/day dehumidifiers as opposed to 1x 300L/day dehumidifier.

CO2 injection

2.60 CO2 injection will usually take place from a single point of entry into the room, however this can often be split out through the use of 4mm t-sections and appropriate tubing, and run above the canopy or along the ceiling.

2.60.1 This tubing can be pierced with a pin or small nail or similar at regular intervals, allowing for the CO2 to "rain down" on the plants below.

2.60.2 The tubing should be run down the length of the drain tables, where the canopy will be.

2.60.3 A "goof plug" may be used to plug and seal the end of the 4mm CO2 lines, so that the CO2 only comes out of the small holes.

2.61 A software “median” sensor (for example, a Home Assistant template sensor) may need to be employed that takes the median CO2 reading across the growspace. This can control a relay which uses a set point (usually a ppm value matched to the maximum PAR for the grow space) to turn a CO2 solenoid on or off. See the Chill Division Automations Guide for setup detail.

2.61.1 The CO2 solenoid should be placed as close as possible to the room for injection. This is because with longer tubing runs, the tube will effectively be pressurized with CO2 that will continue to be introduced to the grow room even after the solenoid is turned off. As such if there is a long run of tubing after the solenoid, but before the grow room, CO2 will continue to seep out even after the desired setpoint has been reached, decreasing the overall accuracy of CO2 injection.

2.62 An automatic CO2 bottle bank-switching manifold should be utilised such as one from Biotek Engineering. This will ensure that CO2 never runs out even when a bottle is empty. They also have the added benefit of requiring no power for the change-over, instead basing the bottle selection based on pressure from the full tank.

2.63 CO2 bottles must not be stored inside the grow room itself. They should instead be located in the larger open factory / warehouse space, or otherwise outside the building in the open air.

2.63.1 Bottles should be secured upright and restrained against falling, regardless of location.

2.63.2 Where bottles are stored inside (such as in the open factory space), the area must have an ambient CO2 sensor (such as an SCD4x-based unit) with alerting, so that a leaking bottle is detected before staff enter the area. The sensor should trigger a remote alert to responsible persons on elevated readings.

2.63.2.1 CO2 is heavier than air and will pool at floor level, so consideration should be given to ventilation of any enclosed area, along with a low-level exhaust.

2.63.3 Where bottles are stored outside in the open air, no leak sensor is required, as any leaking CO2 will simply dissipate.

Lighting

2.63 Pure LED grow lighting is recommended for photosynthetic lighting arrangements. Although some experienced growers may opt for mixed 50/50 lighting of LED<>HPS ratios, it is not suggested for new builds until the grower has more experience with the unique opportunities and challenges it presents, as well as integration of HPS controls into an automation system.

2.64 LED grow lighting should be primarily selected based on both CRI (Color Rendering Index) and diode efficacy.

2.64.1 Although some spectrum of lighting such as blurple has historically been viewed as “all the plant needs”, it makes it significantly harder for a skilled cultivator to observe what is taking place with the plants, early observation of nutrient deficiency or toxicity, and pests or bugs. As such a more balanced spectrum is usually beneficial.

2.64.2 A CRI of 85+ should be sought for LED lighting fixtures, as opposed to the off-balance CRI of ~20-25 for most HPS lights.

2.65 LED grow lighting should be hung sufficiently above the canopy such that there will be minimal issues from radiant heat. Ideally this would be >450mm above where the top-most buds will be.

2.66 A PAR analysis should take place to ensure a consistent hang-distance between fixtures. The optimal lighting hanging height vs distance from the next fixture will vary for every individual fixture, depending on factors such as diode type, bar layout, diode efficacy, lensing and other waterproofing, and physical footprint of the fixture.

2.66.1 When hung optimally, there should be minimal “drop off” or “hot spots” between fixtures.

2.66.2 As a general guide, a 1000w LED should be suitable for approx a 1.2*1.2 up to a 1.5*1.5 area.

2.66.3 Alternatively a 750w LED could be used for a 1.0*1.0 up to a 1.2*1.2 area, again depending on the fixture, any diode lenses etc

2.67 Consideration should be given to upper-canopy vs lower-canopy PAR levels. To calculate the Photosynthetically Active Radiation (PAR) from a lighting fixture at different distances, you can use the inverse square law. This law states that the intensity of light is inversely proportional to the square of the distance from the light source.

2.67.1 The formula to calculate the PAR at a second distance (usually lower-canopy) based on a measurement at a first distance (canopy top) is:
$PAR_2 = PAR_1 \times (distance_1 / distance_2)^2$

2.67.2 For example, with a PAR_1 value of 1200μmol/m²/s at 45cm from the lighting fixture, to calculate the PAR_2 value at 80cm you would calculate `PAR₂ = 1200 × (45/80)²`, which is `1200 × (0.5625)²`, or `1200 × 0.3164`, giving a PAR_2 value of 380μmol/m²/s.

2.67.3 Similarly, with the same PAR_1 of 1200μmol/m²/s measured at 60cm, the lower-canopy PAR_2 at 95cm would be `1200 × (60/95)²` = 479μmol/m²/s. This shows why hanging LEDs at a higher level and running at a higher power output can be beneficial for canopy penetration.

2.68 Additional consideration should be given to any movable isles / tables where the canopy may be, so that the PAR is evenly distributed regardless of where the tables are left situated.

2.69 All fixtures should be tested prior to committing to a whole room fit-out, to ensure that any 0-10v ADC or other PWM signal control is compatible with the automation system.

2.69.1 Although lights may be individually controlled, usually a “whole of room” daisy-chain is sufficient, as individual lights are almost never dimmed independently across a consistent and even canopy.

Room lighting

2.70 Additional room lighting may be installed above the LED grow lights, or in areas of the room which may not be directly illuminated by the LED grow lights.

2.71 Care must be taken to select smart-lighting that can be remotely controlled alongside the LED grow lights. This may take the form of Philips Hue LEDs, or, a Zemismart Zigbee smart-relay light switch.

2.71.1 Color changing LEDs like Philips Hue may be beneficial over a smart-relay light switch for including additional effects, such as changing the color to red and pulsing on/off for alerts when CO2 levels are too high.

2.71.2 Failsafe and other fallback alerts should be set up to ensure that the lighting does not interfere with the photoperiod for the plants, including having the capacity to "start off" after a power outage.

Undercanopy lighting

2.72 Undercanopy lighting should be employed to aid in meeting the European Pharmacopoeia 11.5th Edition Cannabis Flower (Cannabis flos) monograph requirements, whereby the stated label claim must not be exceeded by +/- 10%. Having LED grow lights specifically set under the canopy aiming upwards at the lower parts of the plant can help to minimize the PAR spread from canopy top to bottom. This has a very direct impact on a plants' ability to reach peak cannabinoid levels.

2.73 As with over-canopy LED grow lighting, priority should be given to the CRI. Although red spectrum and far-red is more efficient as canopy penetration, an off-balance CRI can cause issues for the growers observing the plants with diagnosing any issues that may arise.

2.74 All undercanopy LED bars should be dimmable and able to be daisy-chained. This is due to the lighting needing to be at a lower PAR level when initially employed during the second week of the flowering photoperiod.

2.74.1 Where they cannot be dimmed, a single relay to control each tables LEDs may be employed as a secondary measure.

2.74.2 Daisy chaining allows for a single plug to control all of a tables' undercanopy LEDs, which is important when the lights are moving with the tables.

2.75 Stands should be installed for the undercanopy LED bars to keep them slightly elevated off the drain trays, and not directly touching any coco pots or rockwool cubes.

2.76 Most undercanopy fixtures are sufficiently powerful at 100w -> 150w when raised ~350mm off the drain tray.

Tables, isles and plant density

2.77 Moveable tables that hold the drain-trays are often preferred over fixed tables / isles, in order to maximize the amount of canopy space per-room.

2.77.1 This may not be the case where mains-power supply (Such as a 3-phase 60A capacity) is more restrictive than physical floorspace footprint.

2.78 Consideration should be given to plant access when determining table width. A 1.5m wide drain tray is easily divisible by 3x plants each having a 50cm x 50cm space, however this places the middle plants at a 75cm distance from the isle which will greatly reduce the growers ability to interact with and defoliate the plant.

2.78.1 For this reason a 1.2m wide tray should be considered the upper limit where possible.

2.78.2 Trays of 60cm, 90cm or 100cm width can also be very good options, depending on the plant density. These will all divide nicely into common plant density options being every 30cm x 30cm, 45cm x 45cm, 50cm x 50cm and 60cm x 60cm, providing options for the grower to suit their style and the plant needs.

2.78.2.1 A plant density of ~330x330 is the upper-limit of density, 9 plants per-sqm. 250x250 (16 per-sqm) is not recommended for new growers due to the fast time between vegetative growth and flowering (usually <1 week).

2.78.2.2 A plant density of 500x500 is the lower recommended limit of plant density being 4 plants per-sqm. 600x600 (3 per-sqm) is not recommended due to the size of the plants being used for the substrate, a higher density is almost always a better option.

2.79 Tables should be placed in such a way that a grower can access them from all four sides at all times.

2.79.1 If the tables are moveable then permanent access to all sides is not necessary, e.g. one face of a table is up against a wall when moved to one extreme.

2.79.2 Isles and walk spaces should be sized as per §2.2.1 (minimum 800mm, 1200mm preferable).

2.80 Once the table width has been determined, drain trays can then be ordered to match the desired width / length.

2.80.1 Shorter drain trays may be connected to each other lengthways, however a single full length table is preferable.

3. Irrigation system

The irrigation system is expected to feed mostly out of Batch Tanks. The Batch Tank will feed into a Pump. The pump will then go into a T-section controlled by two solenoids. One will recirculate back past a dosing manifold and into the main Reservoir. The other will go through a multi-stage filter and then on to the flower rooms for fertigation.

3.1 Mains water supply may need a Pressure Reducer such as a 32x25mm High flow 345 kpa (PRV3225HF-50) to reduce pressure to ~3.5 BAR (50psi)

3.2 All freshwater should be filtered with a 3-stage + UV filtered or similar equivalent to ensure the water is under 0.1EC.

3.2.1 This includes any borewater, rainwater, tapwater etc

3.2.2 Although RO water is preferable, it can be cost-prohibitive for a small start-up and so a 3-stage + UV is an acceptable substitute that has previously been shown to not cause issues with NZMQS verification.

3.2.3 Filters should be installed in 20 micron, 5 micron, and 1 micron filtration order, with a UV filtration last.

3.3 There should be regular pressure gauges throughout the irrigation system, both pre-filtration, post filtration, throughout any stock mixing, lines to rooms, and on all runs of lateral tubing. Supplementary adaptors may be required to integrate into uPVC / pressure pipe systems.

3.4 All threads must be installed with thread seal tape. All press fittings should use primer and PVC cement.

Nutrient storage area

3.5 The nutrient storage area must be a secure, cool, dry, and well-ventilated space, completely isolated from direct sunlight or intense grow lights.

3.6 Nutrients must be stored in their original, clearly labeled, and tightly sealed containers on sturdy shelves that keep them off the floor.

3.7 A strict first-in, first-out inventory system must be implemented to ensure older stock is used first, preventing degradation.

3.8 It is absolutely critical to store different types of nutrients separately to prevent dangerous chemical reactions. Specifically, acids, bases, and oxidizers must be physically separated from each other using separate cabinets or secondary containment bins.

3.9 The area should be clearly marked with hazard signs, and a complete inventory list along with all relevant Safety Data Sheets (SDS) must be kept in an accessible location nearby.

3.10 A spill containment kit should be readily available in the area.

Stock mixing area

3.11 The stock mixing area should be a dedicated, self-contained space designed with safety and cleanliness as top priorities.

3.12 Consideration for drainage in the area should be given, such that it can be washed with a pressure washer in event of any spills.

3.13 It requires a level, non-slip floor surface for mixing and pouring nutrients to prevent spills.

3.14 The area must have immediate and easy access to a 3-stage + UV filtered water source and a large utility sink.

3.15 All surfaces, including walls and countertops, must be non-porous and resistant to chemical corrosion for easy cleaning.

3.16 The layout should provide ample space for organizing and using dedicated measurement tools for each nutrient component to avoid any chance of cross-contamination.

3.17 This area must also be well-ventilated, preferably with its own exhaust fan to remove any fumes, and it should be equipped with safety essentials like a spill containment kit.

Batch mixing (automatic)

It is expected that the use of a Chill Division Auto Batch Doser system will be implemented. Alternatives may include Bluelab Pro Controller WiFi units, or Dosatron Hobby Cultivator kits.

These instructions were written with the intention of Athena being utilized, with one pump for Core/Fade, one for Grow/Bloom, one for Balance/pH-up, and one for Cleanse/HOCl. It can easily be amended to work with the likes of a 3-part or 4-part base such as Front Row, by daisy-chaining multiple units together.

3.18 Automatic batch tank mixing is expected due to the repetitive nature of regularly creating batch tanks for fertigation. It also removes a lot of the human element of variation.

3.19 A 240L Wheelie Bin is the most functional and cost-effective batch tank / reservoir. Assuming it will never run completely dry, and that a sufficient fall from recirculation is configured to mix the nutrients, these will allow for approx 150L per-refill.

3.20 30mm Pressure Pipe should be run from the main Batch Tank through to the main pump.

3.21 25mm Pressure Pipe should be used out from the pump, both for recirculation but also to the flower rooms.

3.22 A BJZ037 pump with controller should be used for irrigation when anything more than a dozen plants will be fertigated. For rooms where there will be less than two dozen plants (such as a mother / nursery area), a smaller constant pressure pump may be used.

3.23 A 4-port Manifold with Swivel Fittings should be obtained for the base use of a 2-part + pH-up + line cleanse nutrient solution.

3.24 An additional single port should also be expected to be used as a way of draining out the solution from the tank in case of an issue.

3.24.1 For this, a Male to Male Nipple adaptor should be obtained, along with a 25mm Dura Manifold Cap to be installed and keep it covered when it is not needed.

3.25 For each of the ports to be used with a peristaltic pump, a 25mm -> 10mm reducing bush should be installed on the manifold.

3.25.1 Each of the reducing bushes should then have a P1668 male press-fit connector that will then be used to connect the hose to.

3.25.2 A short connection of hose should be used (such as 5cm), into a FFCVM10 check valve, to prevent any fluid from being pushed back into the peristaltic pump.

3.26 A minimum of 3x 25mm PGV Solenoid Valves with Flow Control should be used. One for fresh (filtered) water in. The second to control recirculation to the manifold and back into the reservoir. The last solenoid will control fertigation to the grow room. By adjusting which solenoid is open, the pump should automatically spin up when it notices the pressure drop, and allow for batch tank refilling and fertigation all off the one pump.

3.26.1 A 4-channel relay should be used to control each solenoid individually.

3.26.2 Additional solenoids may optionally be fitted closer to each individual room / table.

3.26.3 A suitable 24v AC power supply must be obtained that will power the solenoids, and extra care taken to ensure DC voltage is not used. One of the specified 150VA power supplies should safely handle 10x of the Hunter PGV solenoid valves and not exceed 80% capacity.

3.27 The Reservoir should be fitted with an Ultrasonic Sensor on the lid, to ascertain the levels of remaining batch nutrients, and for triggering automatic refills.

3.27.1 This must be an Ultrasonic sensor, not a Time-of-Flight (ToF) sensor, due to the ToF being a laser that goes through the water giving inaccurate readings.

Fertigation system / plant-feeding

This section covers fertigation and the application of fertilized nutrient feed to plants, specifically for flower, but is repeatable in vege / mothering areas.

This section also assumes the build-out of the Batch mixing (automatic) has been followed.

3.28 Immediately after the solenoid that will control fertigation, there should be a Barrel Union, as this will allow for easily removal / replacement of the whole filter housings.

3.29 Two filters are the minimum, 3x is recommended.

3.29.1 Where two filters are used, they should be 300 micron and 130 micron.

3.29.2 Where 3x are used, they should be 300 micron, 130 micron and 80 micron. A larger mesh unit may need to be purchased and the filter obtained separately whereby an 80 micron unit is not readily available.

3.30 25mm Pressure Pipe should be run the remainder of the way to any flower rooms.

3.31 A Pressure Gauge should be installed in the flower room, before the Pressure Pipe is tee'd off to go to any tables.

3.32 Where flexible tubing is required to run to moving tray tables, it should be tee'd off from the main fertigation, a 20psi Pressure Reducing Adapter installed, followed by a Check Valve to ensure no backflow, and then Washing Machine Inlet Hose is used for its flexibility and braided sheathing.

3.32.1 The Pressure Reducing Adapter should be 20psi, as the Netafim PCJs only require 10psi to operate.

3.33 After the Washing Machine Inlet Hose, the table must be run with Lateral Tubing. The Lateral Tubing is the only tubing that is suitable for installation of PCJs (Pressure Compensating Drippers). This is due to the material self-healing after being punctured.

3.34 Ratchet clamps should be used on all lateral tubing. These may require more than just "hand clamp" force when installing, and appropriate tools sourced to assist with securing them such that they will not pop off when under pressure of up to 3BAR.

3.35 Where more than one line of Lateral Tubing will be run, such as down both sides of a 1.5m wide table, 19mm tee and 19mm elbow sections should be used, secured with ratchet clamps.

3.36 The Lateral Tubing should then be punctured at intervals with a Punch Tool, anywhere that a plant will be placed.

3.36.1 A clamping Punch Tool should be used as opposed to a needle-style, as a clamping tool will hold the tubing in-place more firmly. This provides a full and clean piercing of the tubing such that the tubing can easily "heal" around the PCJ once inserted.

3.37 Netafim Pressure Compensating Drippers (PCJs) should then be installed in the respective holes for each plant.

3.37.1 Netafim PCJs should be used for their self-cleaning and reliability, as they are less susceptible to clogging compared with others.

3.37.2 If a Rockwool substrate is being utilized, 2x 2L/hr drippers per-cube should be used, and attached to barbs / stakes. Pre-assembled drippers may be used for convenience.

3.37.2.1 Alternatively 2x 1.2L/hr may be used, however the fertigation lines should have 3x filters down to 80 micron if this is used.

3.37.3 If a Coco substrate is being utilized, 1x 4L/hr dripper per-bag should be installed and attached to 120mm Netbows (which each have 4x smaller dripper holes).

3.37.4 The PCJs should be attached to the barbs / netbows with 3mm tubing.

3.38 There should be at least one pressure gauge used per-table, usually installed near the end of the line. Operators may additionally want to install one closer to the start of the table, before any tee-joins, to be able to compare pressure across the entire run and ensure uniformity.

3.39 At the end of each run of Lateral Tubing, a 19mm Valve should be installed to assist with manually purging lines.

Drainage

3.40 Drainage and the flow of waste should be carefully considered, with the expectation that a leak may occur from anywhere and needs to be able to be accessed easily to repair.

3.41 In-floor drainage is optimal when constructing a drainage system. As noted in §2.5, an elevated / secondary floor makes this easier to achieve.

3.42 Where in-floor drainage is not available, the run-off from tables should be fed into a Sump Pump or similar system and pumped into a small reservoir. This is because the head height on most sump pump systems is not going to be able to handle moving any volume of water very far.

3.43 A secondary reservoir and larger waste pump may need to be used to further pump any runoff into waste.

3.44 All wastewater and runoff must ultimately discharge to an approved point. Trade-waste consent should be sought from the relevant network operator (such as Watercare in Auckland, or the local council / water authority elsewhere) before discharging nutrient-laden runoff to the sewer, as untreated horticultural runoff can breach trade-waste limits on nutrients, pH, and suspended solids.

3.44.1 Where a sump / waste pump is used (per items 42–43), size the collection reservoir and pump for the peak runoff volume of a full fertigation event, plus a margin for a simultaneous leak or wash-down.

3.44.2 Consideration should be given to how runoff EC / nutrient concentration may need to be managed, diluted, or dosed to meet trade-waste consent conditions.

3.44.3 Seek advice early from the network operator and a drainlayer, as consent conditions can influence the drainage design and the size of any reservoirs / pumps required.

4. Nursery / mothering / cloning

A dedicated and isolated nursery area is the foundation of a clean and productive cultivation facility. It is vital for maintaining healthy, vigorous mother plants and ensuring a pest-free start for every crop cycle. This space serves as both a secure quarantine zone for any new genetics being introduced and as a stable propagation environment for creating healthy clones.

4.1 The nursery can be a small, fully self-contained room or a dedicated tent within a larger area. Regardless of its form, it must be completely sealed from other grow spaces to prevent any potential for cross-contamination of pests or pathogens.

4.2 An antechamber or separate entrance is highly recommended.

4.3 Environmental controls in the nursery must be precise and independent from other cultivation areas. It requires its own dedicated systems for lighting, ventilation, and humidity to maintain the specific, gentle conditions needed for rooting clones and caring for mother plants, which typically involve lower light intensity and higher humidity than flowering rooms.

4.4 All surfaces within the nursery, including walls, floors, and benches, must be made of non-porous materials such as FRP panels or refrigeration panels. This allows for frequent and thorough cleaning and sanitization between propagation cycles, which is essential for preventing disease outbreaks.

4.5 The area should be thoughtfully designed to house mother plants in a permanent vegetative state, allowing them ample space to grow without stress.

4.6 It also needs dedicated, well-organized space for clone domes and trays (Or a VPDome equivalent), equipped with appropriate low-intensity LED bar lighting and thermostatically controlled heating mats to encourage consistent and successful rooting.

> **4.6.1** Ensure the VPDome can fit Grodan Gro-Smart trays, as these are recommended for cloning, being 20” or 51cm lengthways and half that on the short side.

5. Drying area

The drying area is a critical environment where the quality, aroma, and preservation of the final product are determined. This space must be designed for absolute and precise control over temperature and humidity to ensure a slow, even, and gentle drying process, which is essential for preserving terpenes and preventing mold.

5.1 The room must be completely sealed, light-proof, and equipped with its own dedicated, and preferably redundant, air conditioning and dehumidification units. These systems must be capable of maintaining a consistent temperature of approximately 16 degrees Celsius and a relative humidity of 60-65 percent without significant fluctuations.

5.2 Air circulation is important for preventing mold, but it must be gentle. No fans should ever blow directly onto the hanging plants as this will cause them to dry too quickly. The ventilation system should be designed to create slow, gentle, and indirect airflow throughout the entire room, ensuring a consistent environment for all plants.

5.3 The room should be equipped with an efficient system for hanging plants, such as stainless steel wires or rolling racks. These systems should be made of non-porous materials like plastic or metal to prevent microbial contamination and allow for easy cleaning.

5.3.1 Scaffolding pipe that has been well cleaned and sterilized may be used, along with S-hooks to hang a whole-plant.

5.3.2 Metal coathangars that fit over a scaffolding pipe may alternatively be utilized for part-plant hanging where GMP Chain of Custody requirements permit pre-bucked flower delivery.

5.4 All surfaces in the drying area, including walls, floors, and ceilings, must be non-porous and easy to clean and sanitize. A strict protocol of thoroughly cleaning and sterilizing the entire room with Peracetic Acid between each harvest is mandatory.

5.5 Sizing a drying room can be difficult, but it should never need to be larger than a single flower-room. Plants can often be hung in a multi-tiered solution, however movement around the plants for staff needs to be considered, coupled with the ability for air to be moved around the plants without directly blowing on them, and humidity removed from the room.

5.6 As a rule of thumb, single-tier drying (plants hung in a single layer) needs roughly the same floor area as the flowering canopy it serves — the industry guidance is that whole plants take up much the same space drying as they did growing, as they do not shrink significantly when cut. Where plants are hung in a multi-tiered / stacked configuration, this can be reduced to approximately half of the flowering canopy area, or less. The 20sqm drying allocation in the §2 floorplan estimates assumes such a multi-tiered layout serving ~40-50sqm of canopy; if single-tier drying is preferred, allocate correspondingly more space.

6. Data network

A reliable data network is an absolute necessity, with local control of all devices to avoid the possibility of any external interference from “the cloud”. Delays or missed actions on things such as irrigation events or CO2 injection can be catastrophic both for plants and personnel. A locally controlled system has fewer points of failure and should be viewed as the gold standard to be utilized. Similarly, cabled-in sensors are more reliable than wireless where external signal interference may cause issues, and PoE powered devices should be the priority where possible.

6.1 The network should be designed such that an internet outage does not cause automations and other such devices to stop working.

6.1.1 This can be tested by removing the ethernet cable between the router and the fibre ONT.

6.1.2 It should also be tested by removing the fibre cable from the ONT to ensure that it’s not just limited to working when it notices there’s nothing plugged into the WAN.

6.2 The router should be a stand-alone router device only, such as a Ubiquiti Edgerouter-X, and should not be used as a WiFi router. This permits better placement of WiFi access points, while also avoiding most common ISP-supplied routers that are prone to being unreliable.

6.2.1 Many other routers / brands also have reliability issues. If in doubt, use the recommended suggestion. All cameras, IoT devices, AC units, fileserver / DVR, and more, will all contribute to being “active devices” on the network, which many routers are simply not equipped to handle and will cause problems for the operator.

6.3 WiFi should be a Ubiquiti U7-Lite, U6-Lite or similar. Although these units won’t be used for their capacity to handle a significant number of concurrently connected devices, it will be used specifically for its reliability. They also have high powered 2.4Ghz which is what almost all IoT devices will be using to connect where PoE is unavailable.

6.4 Depending on placement, a single U7-Lite unit per growroom should be used.

6.4.1 For more compact facilities where the floorplan permits, a more centralized U7-LR (Long Range) may be used, but consideration should be given to the physical location of the device ensuring it is as central as possible.

6.5 A 48-port network switch such as the ES-48-500W should be obtained for the primary network switch.

6.5.1 Even a small facility will likely exceed a 24-port switch capacity such as a US-24-250W, with a dozen cameras, DVR, WiFi access points, EdgeRouter, a larger 48-port is the next size up and provides room for expansion.

6.5.2 Alternatively two large switches like this can be used with a small uplink cable between them.

6.6 Where practical, the data network may be designed in a "starburst" shape. The primary router will be attached to the main network switch, along with most central "server" devices. This main network switch then connected to other secondary switches such as 8-port or 16-port switches closer to each of the growrooms. This design allows for a single "uplink" cable to be run to the smaller network switches, and then shorter cable runs to cameras, WiFi access points, and IoT devices, saving on cost and time.

6.7 Ethernet cabling can be run as Cat5e or Cat6. Any runs that are longer than ~100m will need an interim network switch in between them, to overcome the limits defined in ISO/IEC 11801.

6.7.1 Due to the nature of the devices likely to be running on the network, such as IoT devices operating at speeds that can be measured in "bps", there is not likely to be a need for anything faster than 10/100 Fast Ethernet except for between servers on the primary / central network switch.

7. Electrical wiring

The electrical system in a cultivation facility must be designed and installed with safety, reliability, and future expansion in mind. All work must be completed by a qualified and licensed electrician to the highest commercial standards, accounting for the unique high-load and high-humidity environment of a grow facility.

7.1 All electrical outlets and fixtures inside the grow rooms must be installed on the ceiling or as high on the walls as possible. This strategic placement minimizes the risk of contact with water during irrigation, foliar spraying, or room cleaning.

7.2 All power outlets and fittings should be waterproof (IP65 rated or higher).

7.3 High-load equipment, particularly lighting arrays and dehumidifiers, should be wired to their own dedicated circuits or split across multiple circuits, with appropriate amperage ratings to prevent overloads and ensure stable, continuous operation.

7.4 The total electrical load for each room must be meticulously calculated by the electrician. This calculation must account for the absolute peak demand when all equipment is running simultaneously. The electrical panel and all wiring must be specified to handle this maximum load safely and continuously, with a buffer of at least 20% being desirable for safety and future upgrades. The load should be broken down as follows:

7.4.1 **LED Lighting:** These represent the largest and most consistent electrical load. The circuits must be designed to handle the full wattage of all lights running for 18 or more hours continuously.

7.4.1.1 Also consider how power cables will be routed to drain-tables with undercanopy lighting.

7.4.2 **Climate Control**: Dehumidifiers and air conditioners have very high power requirements, especially the large inrush current they draw upon startup. Each of these units requires its own dedicated circuit appropriately sized to handle this initial surge without tripping breakers.

7.4.3 **Air Circulation and Ventilation**: This includes all inline fans for air scrubbing and exhaust, as well as multiple oscillating fans within the room. While the individual draw of each fan is relatively small, their collective and continuous operation contributes significantly to the overall load and must be factored into the circuit design.

7.4.3.1 Spacing and placement of fans should be considered as part of the room design, but additional and surplus power plugs should be added in strategic locations to ensure that it can easily be expanded upon once the room is in-use.

7.4.4 **Ancillary Equipment:** This category includes devices like humidifiers, irrigation pumps, and CO2 solenoids. Their power draw is typically low, but they must be included in the total load calculation to ensure the electrical system is not overloaded.

7.5 It is critical that the electrician balances the various equipment loads evenly across all three phases. Proper load balancing prevents overheating, improves efficiency, and ensures the long-term stability and safety of the entire electrical infrastructure.

Changelog

- Draft 4 - 20260703 (in progress)

- Reworded wall / ceiling lining expectations under §2: insulated chiller panels (rockwool core, aluminium or Colorsteel skins) are now the base expectation, with ACM board as an acceptable secondary option installed over a separately insulated build-up
- Added §1.23 requiring floorplans to label each room's purpose, with matching physical labels on rooms, and free-standing filter locations noted on the floorplan (renumbered Physical Security Arrangements accordingly)
- Added door-state sensor requirement under Physical Security Arrangements (Zigbee contact sensors as the common choice; hall-effect / limit-switch wired options as alternatives), including roller door state sensing
- Added baseline camera hardware requirements: PoE-cabled only (never battery), 1280x720 @ 2fps minimum, continuous 24/7 recording, automatic IR
- Added secure storage room item: standalone controlled-drug safe / vault no longer required; the most secure room (typically drying / processing) is to be designated and noted on the floorplan
- Added loading bay design intent: vehicle reversed flush against the roller door opening to obscure loading / unloading from outside view
- Added Police notification pointer to §1 preamble (full requirement lives in the Security Policies & Procedures)
- Added note that M5Stack ENV.IV is EOL (ENV.III has returned); now referred to generically as "ENV Sensor(s)" throughout
- Added awareness note under §2 Sensors that grow rooms must be fully sealed or negatively pressurized, with specific targets deferred to the Security Policies & Procedures
- Added §2 CO2 injection item: bottles must not be stored in the grow room — open factory space (with ambient CO2 sensor + alerting) or outside in open air (no sensor required)
- RFC 2119 consistency pass: removed bold / caps styling from keywords, replaced "shall" with "must" throughout, fixed descriptive phrasing that was intended as requirements (siren, PIN lockouts), and resolved the "must … where possible" contradiction for cabled sensors
- Unified the 120m residential distance as a strong recommendation (should) in both §1.1 and §1.1.4.1, with explicit case-by-case review / Agency discretion wording
- Duplicates cleanup: removed the triplicated internal carbon-scrubbing requirement (kept §1.18.1 + §2 Air scrubbing); trimmed the §2.7–8 silicon / door sealing to cross-reference §1.19–20 while keeping grow-room-specific detail; made aisle-width (§2 Tables) and plant-access (§2.2.2) items cross-reference their canonical statements; linked §3 Drainage 41 back to §2.5 for elevated flooring

- Unclear cleanup: filled the §3.44 drainage placeholder with sump / waste sizing and trade-waste consent (e.g. Watercare) guidance; reconciled drying-area sizing (§5.6) to the industry standard — single-tier ≈ 1:1 with canopy, multi-tier ≈ half, and clarified the §2 20sqm figure assumes multi-tier; fixed the broken PAR inverse-square-law superscripts and stated PAR_1 in both worked examples; reworded the "templated CO2 sensor" as a software / Home Assistant median sensor with a pointer to the Automations Guide; clarified §2.10.3 that anchoring panels to the roof makes them a permanent building fixture likely needing building consent
- Formatting normalization for publishing: removed the stale embedded table of contents (the site generates its own) and the two empty trailing headings; stripped Google-Docs `{#anchor}` heading attributes; promoted the chapter titles from `N. # Title` list-items to proper `# N. Title` headings; and normalized inconsistent list indentation so numbering renders correctly as §chapter.item (e.g. §1.33, §2.63). No wording changed in this step.
- RFC 2119: cited the Semantic Versioning 2.0.0 specification (semver.org) and clarified ISO 8601 as the date-based option in §1.21
- Draft 3 - 20250904
- Added Duress to Physical Security Arrangements
- Added information about Netafim PCJs specifically due to being less susceptible to clogging
- Added 3-stage + UV filtration order to irrigation filtration
- Added measurements for Gro-Smart trays for VPDome
- Added clarification of Utlrasonic for Res vs ToF sensor
- Added estimates for dividing up a new warehouse floorplan under §2, though it could probably do with a little further refinement as time goes on
- Added details under §2 for chiller panel materials, room span, fire / safety details
- Draft 2 - 20250821
- Changed explanation for dehumidification requirements
- Added link to Humidifier recommendation, clarified position/orientation
- Added information about AC Infinity fan controllers for ventilation
- Added an initial guide to drying area sizing, may need further refinement
- Added Ravenseal links
- Fixed explanation for insulation
- Fixed explanation for raised flooring for drainage
- Changed Air Conditioning to add more info about placement / control
- Changed explanation for PAR levels in Lighting, now recommending 1000w
- Added links to room lighting and added explanation for Hue over Zemismart relays
- Added example weatherproof power socket to §7 - Electrical
- Added explanation for HEPA / MERV filters impacting microbial levels
- Added relay details to the irrigation solenoid section for Batch mixing (automatic)

- Added link to 3-stage + UV filtration system in Irrigation section
- Changed Nutrient Mixing, this now comes earlier in the Irrigation section to allow for better flow between subsections
- Added Fertigation system / plant-feeding section
- Added Drainage section